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PROCEEDINGS

Congenital radial head dislocation associated with radial procurvatum - Case report

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INTRODUCTION

Congenital dislocation of the radial head is a rare condition affecting one or both forearms of young chondrodystrophic dogs. No gender predisposition has been described and the etiology remains to be determined, although genetic factors are thought to be involved. Clinical signs may be present at birth, but are more often noted from the third month of age. Major clinical features are lameness associated with outward projection of the elbow, pain on joint manipulation and reduced range of motion. There may be concurrent, moderate to severe radial deformity with shortening of the ulna in response to osteoarthritis (OA) and abnormal growth, particularly at the distal ulnar physis. Premature closure of the distal ulnar physis is the major cause of impaired radial growth. Limited longitudinal ulnar growth may lead to excessive radial procurvatum, with outward antebrachial rotation and carpal valgus. This study describes a case of congenital radial head dislocation associated with marked radial procurvatum. Different surgical modalities were combined to achieve correction of orthopedic deformities. Ambulation function was successfully restored.

MATERIALS AND METHODS

An intact male Shih-tzu dog aged 6 months and weighing 6.3 kg was referred with a history of mild progressive right thoracic limb lameness and elbow deformity evident from the age of three months. Subjective gait assessment revealed severe right thoracic limb lameness at the walk. A firm swelling was noted at the lateral aspect of the right elbow. Passive elbow range of motion was painful and reduced. Mild carpal valgus was also detected. Radiographic assessment of the left and right forearms was performed, including both, the elbow and carpal joints. Orthogonal projections revealed caudolateral dislocation of the radial head with preserved humero-ulnar joint space in the right thoracic limb and bilateral radial procurvatum, worse on the right side. Proximal radial growth plates were open and had normal radiographic appearance; irregularities were noted at the level of the distal physis of the right ulna. Computed tomography of the right forearm revealed loss of articular congruence between the radial head and the humeral condyle, and mild irregularities on the surface of the lateral epicondyle of the humerus; changes were consistent with angular/rotational radial deformity associated with humeroradial and caudal radial head dislocation. The CORA (center of rotation of angulation) method was used for procurvatum quantification (39 degrees) based on joint angles of normal dogs of the same breed. Surgical treatment was then planned to correct the radial head dislocation and excessive radial procurvatum, as dogs naturally have some radial procurvatum. Surgical procedures performed to achieve humero-radio-ulnar joint stabilization and procurvatum correction were proximal ulnar osteotomy, cranial closing radial wedge osteotomy and internal fixation with 2.0 mm 5 hole locking reconstruction plate, anchor fixation to the lateral humeral condyle and cerclage wiring at the lateral aspect of the radial head.

RESULTS

Correction of outward forearm deviation and restored elbow flexion-extension motion were evident immediately after surgery. Weight bearing was resumed within 10 days of surgery; at that time there was radiographic evidence of preserved articular congruence and procurvatum correction; physical therapy was then started. Eight-week follow-up revealed mild lameness and increased, pain-free passive elbow range of motion. Post-operative radiographic findings within 8 weeks of surgery were as follows: healed radial osteotomy, periosteal reaction at the ulnar osteotomy site and broken cerclage wire. Six months after surgery (last follow-up) the dogs was walking normally and showed no pain on right thoracic limb manipulation. Cerclage wiring failure seems therefore to have had no major consequences.

CONCLUSION

Case reports describing congenital radial head dislocation in dogs are often inconsistent and fail to provide

sufficient support for decision making in clinical practice. In this case, tomographic assessment was vital for deformity quantification and surgical procedures proved effective for correction of congenital elbow dislocation and excessive radial procurvatum, with no major complications. Antebrachial realignment was successfully achieved and normal, pain-free ambulation restored.

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